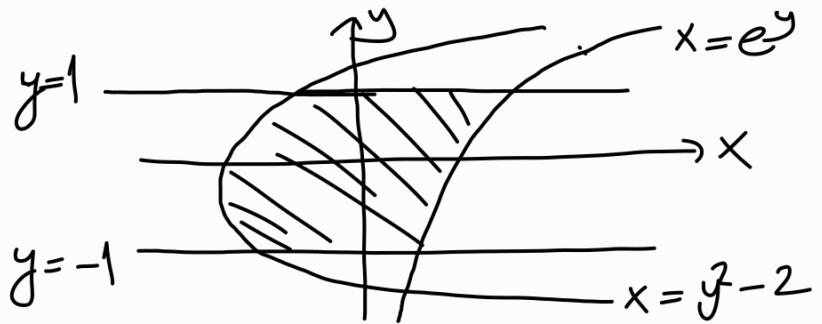
Ex.: Find the area of the region bounded by the curves $y=\sin x$, $y=\cos x$, $x=0$, and $x=\frac{\pi}{2}$

Ex.: Find the area of the shaded region



<u>Ex.:</u> Use calculus to find the area of the triangle with the given vertices: $(0,0)$, $(2,1)$, $(-1,6)$.

Ex.: Find the volume of the solid obtained by rotating the region bounded by the given curves about the specified line. (Disk or Washer)

a) $y = x^3$, $y = x$, $x \geq 0$; about the x-axis

b) $y^2 = x$, $x = 2y$; about the y-axis

c) $y = 1 + \sec x$, $y = 3$; about $y = 1$

Ex.: Use the method of cylindrical shells to find the volume generated by rotating the region bounded by the given curves about the specified line.

a) $y=\frac{1}{x}$, $y=0$, $x=1$, $x=2$; about y-axis

b) $y=e^{-x^2}$, $y=0$, $x=0$, $x=1$; about y-axis

c) $y=\ln x$, $y=0$, $x=2$; about y-axis